

HW9

(20. Solving: $av_1 + bv_2 + cv_3 = 0$

$$a \begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix} + b \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix} + c \begin{bmatrix} 1 \\ 5 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{cases} a + 2b + c = 0 \\ -2a - b + 5c = 0 \\ a + 3b = 0 \rightarrow a = -3b \end{cases}$$

$$\rightarrow -3b + 2b + c = 0 \rightarrow -b + c = 0 \rightarrow c = b$$

$$\rightarrow -2(-3b) - b + 5(b) = 0$$

$$\rightarrow 10b = 0 \rightarrow b = 0$$

$a = b = c = 0 \rightarrow$ vectors are linearly independent ✓

(51. $S = \left\{ \begin{bmatrix} 2 \\ -1 \\ 2 \end{bmatrix}, \begin{bmatrix} 3 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}, \begin{bmatrix} 5 \\ -1 \\ 3 \end{bmatrix} \right\}$

$\text{Span}(T) = \text{Span}(S)$

$A = \begin{bmatrix} 2 & 3 & 1 & 5 \\ -1 & 0 & 1 & -1 \\ 2 & 1 & -1 & 3 \end{bmatrix}$

$R_k \leftarrow R_k - \left(\frac{A_{kj}}{A_{ij}} \right) R_i$

* Gaussian Elimination:

Standard procedure so we can start with 1.

1. $\begin{bmatrix} 2 & 3 & 1 & 5 \\ -1 & 0 & 1 & -1 \\ 2 & 1 & -1 & 3 \end{bmatrix}$ Swap row 1 and 2 $\rightarrow \begin{bmatrix} -1 & 0 & 1 & -1 \\ 2 & 3 & 1 & 5 \\ 2 & 1 & -1 & 3 \end{bmatrix}$ Gets 1's at top

2. $R_1(-1) = \begin{bmatrix} 1 & 0 & -1 & 1 \\ 2 & 3 & 1 & 5 \\ 2 & 1 & -1 & 3 \end{bmatrix}$

3. $R_2 \leftarrow R_2 - 2R_1 = \begin{bmatrix} -1 & 0 & 1 & -1 \\ 0 & 3 & 3 & 3 \\ 2 & 1 & -1 & 3 \end{bmatrix}$

4. $R_3 \leftarrow R_3 - 2R_1 = \begin{bmatrix} 1 & 0 & -1 & 1 \\ 0 & 3 & 3 & 3 \\ 0 & 1 & 1 & 1 \end{bmatrix}$

5. $R_2 \leftarrow \frac{1}{3} R_2 = \begin{bmatrix} 1 & 0 & -1 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 \end{bmatrix}$

6. $R_3 \leftarrow R_3 - R_2 = \begin{bmatrix} 1 & 0 & -1 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \rightarrow$ Need a diagonal? ✓

Pivot columns are 1 and 2 as they contain leading 1's in Row Echelon Form. So the first 2 original vectors form a basis for the span of S. $T = \left\{ \begin{bmatrix} 2 \\ -1 \\ 2 \end{bmatrix}, \begin{bmatrix} 3 \\ 0 \\ 1 \end{bmatrix} \right\}$